WORKED SOLUTION

STEP III 1992, Question 2

Pure • Matrix Powers and Induction

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q2

The key idea

Everything rests on one fact: $\mathbf{J}^2 = 2\mathbf{J}$. That means any product of matrices of the form $\mathbf{I} + p\mathbf{J}$ is again of that form, so we only ever have to track a single number. The awkward-looking expressions $\frac{1}{2}[(1+2a)^k - 1]$ become transparent once we notice that it is $1 + 2p$, not p, that behaves nicely under multiplication.

Solution

A multiplication rule

Since

$$\mathbf{J}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = 2\mathbf{J},$$

and $\mathbf{I}$ commutes with everything, for any real p and q

$$(\mathbf{I} + p\mathbf{J})(\mathbf{I} + q\mathbf{J}) = \mathbf{I} + (p+q)\mathbf{J} + pq\mathbf{J}^2 = \mathbf{I} + (p + q + 2pq)\mathbf{J}. \qquad (*)$$

Notice that

$$1 + 2(p + q + 2pq) = (1 + 2p)(1 + 2q),$$

so if we attach the number $1 + 2p$ to the matrix $\mathbf{I} + p\mathbf{J}$, multiplying matrices corresponds to multiplying these numbers. For $\mathbf{A} = \mathbf{I} + a\mathbf{J}$ the attached number is $1 + 2a$.

The squares and cubes

By $(*)$ with $p = q = a$,

$$\mathbf{A}^2 = \mathbf{I} + (2a + 2a^2)\mathbf{J}, \qquad \text{and} \qquad \tfrac{1}{2}[(1+2a)^2 - 1] = \tfrac{1}{2}(4a + 4a^2) = 2a + 2a^2,$$

so $\mathbf{A}^2 = \mathbf{I} + \frac{1}{2}[(1+2a)^2 - 1]\mathbf{J}$, as required.

For the cube, write $\mathbf{A}^3 = \mathbf{A}^2\mathbf{A}$ and use $(*)$ with $p = 2a + 2a^2$, $q = a$:

$$\begin{aligned} p + q + 2pq &= 2a + 2a^2 + a + 2a(2a + 2a^2) \\ &= 3a + 6a^2 + 4a^3 \\ &= \tfrac{1}{2}(6a + 12a^2 + 8a^3) = \tfrac{1}{2}[(1 + 2a)^3 - 1], \end{aligned}$$

using $(1 + 2a)^3 = 1 + 6a + 12a^2 + 8a^3$. So $\mathbf{A}^3 = \mathbf{I} + \frac{1}{2}[(1 + 2a)^3 - 1]\mathbf{J}$, as required.

For the fourth power, the quickest route is $\mathbf{A}^4 = \mathbf{A}^2\mathbf{A}^2$. With $p = q = \frac{1}{2}[(1 + 2a)^2 - 1]$ we have $1 + 2p = (1 + 2a)^2$, so the number attached to $\mathbf{A}^4$ is $(1 + 2a)^2(1 + 2a)^2 = (1 + 2a)^4$. Hence

$$\mathbf{A}^4 = \mathbf{I} + \tfrac{1}{2}[(1 + 2a)^4 - 1]\mathbf{J} = \mathbf{I} + (4a + 12a^2 + 16a^3 + 8a^4)\mathbf{J}.$$

The general power

The pattern suggests

$$p_k = \tfrac{1}{2}[(1 + 2a)^k - 1], \qquad \text{equivalently} \qquad 1 + 2p_k = (1 + 2a)^k.$$

We prove $\mathbf{A}^k = \mathbf{I} + p_k\mathbf{J}$ for all integers $k \geq 1$ by induction.

Base case. For $k = 1$, $p_1 = \frac{1}{2}(1 + 2a - 1) = a$, and $\mathbf{A}^1 = \mathbf{I} + a\mathbf{J}$. (The formula also gives $p_0 = 0$, matching $\mathbf{A}^0 = \mathbf{I}$.)

Inductive step. Suppose $\mathbf{A}^k = \mathbf{I} + p_k\mathbf{J}$ with p_k as above. Then by $(*)$,

$$\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A} = \mathbf{I} + (p_k + a + 2ap_k)\mathbf{J},$$

and

$$\begin{aligned} p_k + a + 2ap_k &= (1 + 2a)p_k + a \\ &= \tfrac{1}{2}(1 + 2a)[(1 + 2a)^k - 1] + a \\ &= \tfrac{1}{2}(1 + 2a)^{k+1} - \tfrac{1}{2}(1 + 2a) + a \\ &= \tfrac{1}{2}[(1 + 2a)^{k+1} - 1] = p_{k+1}. \end{aligned}$$

So $\mathbf{A}^{k+1} = \mathbf{I} + p_{k+1}\mathbf{J}$, and by induction

$$\mathbf{A}^k = \mathbf{I} + \tfrac{1}{2}[(1 + 2a)^k - 1]\mathbf{J} \qquad \text{for all integers } k \geq 1.$$

Ask yourself Where did the guess for p_k come from, if I hadn't been handed $\mathbf{A}^2$ and $\mathbf{A}^3$?

From the recurrence itself: the inductive step shows $p_{k+1} = (1 + 2a)p_k + a$, a first-order linear recurrence. Adding $\frac{1}{2}$ to both sides gives $p_{k+1} + \frac{1}{2} = (1 + 2a)(p_k + \frac{1}{2})$, so $p_k + \frac{1}{2}$ is geometric with ratio $1 + 2a$ and first term $a + \frac{1}{2}$. That is exactly $p_k + \frac{1}{2} = \frac{1}{2}(1 + 2a)^k$.

Negative powers (a bonus). If $a \neq -\frac{1}{2}$ the formula works for $k = -1$ too: $p_{-1} = -\dfrac{a}{1+2a}$, and $(*)$ gives $a + p_{-1} + 2ap_{-1} = a - \dfrac{a(1+2a)}{1+2a} = 0$, so $(\mathbf{I}+a\mathbf{J})(\mathbf{I}+p_{-1}\mathbf{J}) = \mathbf{I}$. If $a = -\frac{1}{2}$ then $\mathbf{A} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$ is singular, and the formula gives $p_k = -\frac{1}{2}$ for every $k \geq 1$: indeed $\mathbf{A}^2 = \mathbf{A}$.

Checks

- Take $a = 1$: $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, and multiplying out gives $\mathbf{A}^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, $\mathbf{A}^4 = \begin{pmatrix} 41 & 40 \\ 40 & 41 \end{pmatrix}$. The formula gives $p_2 = \frac{1}{2}(9-1) = 4$ and $p_4 = \frac{1}{2}(81-1) = 40$.
- $\mathbf{A}$ has eigenvector $(1,1)$ with eigenvalue $1+2a$, and $(1,-1)$ with eigenvalue 1. So $\mathbf{A}^k$ has eigenvalues $(1+2a)^k$ and 1; the matrix $\mathbf{I}+p_k\mathbf{J}$ has eigenvalues $1+2p_k$ and 1, which agree.
- We checked $\mathbf{A}^k = \mathbf{I} + p_k\mathbf{J}$ in exact rational arithmetic for $0 \leq k \leq 12$ and forty random values of a, including $a = -\frac{1}{2}$.

Common slips

- Expanding $(\mathbf{I}+a\mathbf{J})^k$ by the binomial theorem is legitimate (since $\mathbf{I}$ and $\mathbf{J}$ commute) but forgetting $\mathbf{J}^r = 2^{r-1}\mathbf{J}$, not $\mathbf{J}$, gives the wrong coefficients.
- In the inductive step, writing the conclusion for p_{k+1} first and working backwards is not a proof; start from $\mathbf{A}^k\mathbf{A}$.

Beyond STEP

The number $1+2p$ we attached to $\mathbf{I}+p\mathbf{J}$ is the eigenvalue of that matrix on the vector $(1,1)$; every such matrix shares the eigenvectors $(1,1)$ and $(1,-1)$, so they are all diagonalised by the same change of basis, and powers are computed by simply raising eigenvalues to the kth power. This is the idea behind diagonalisation, which turns matrix powers, matrix exponentials and linear recurrences into scalar calculations. Strang's *Introduction to Linear Algebra* develops it with many worked examples.

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